

Lab Report Name:
**Report on Recognition of Need for Software-Based
Monitoring and Control System at Bangladesh Cable Shilpa
Limited**

Introduction :

In today's industrial world, the use of technology and software systems has become very important for maintaining efficiency and productivity. During our visit to **Bangladesh Cable Shilpa Limited**, we observed the working environment and discussed different operational aspects with the manager and employees. The company is involved in the production of various types of cables, which requires proper coordination among different departments such as production, inventory, and quality control.

From our observation, we found that many activities are still handled manually or with limited digital support. Although the system is functional, it is not fully efficient for handling large-scale operations. Manual data recording, delayed communication, and lack of proper monitoring make the process slower and sometimes less reliable. Therefore, there is a clear need for a modern system that can manage all activities in a more organized and automated way. Our proposed system aims to develop a software-based monitoring and control system that can improve overall performance and reduce unnecessary workload.





Problem :

The current system in the industry has several limitations that affect its efficiency. One major problem is the use of manual methods for recording data. Workers often write down production and inventory details by hand, which takes time and increases the chances of mistakes. These errors can create confusion and affect decision-making.

Another important issue is the lack of real-time monitoring. The management cannot instantly check the condition of machines or the progress of production. As a result, if any problem occurs, it takes time to identify and fix it. This delay can reduce productivity and sometimes cause losses.

Communication between different departments is also not smooth. Information is often passed manually, which can lead to misunderstandings or delays. In addition, preparing reports is a time-consuming process because data has to be collected from different sources and then organized.

Energy usage is another concern. Since there is no proper system to track energy consumption, it becomes difficult to control unnecessary usage. All these problems clearly show that the existing system is not sufficient for a modern industrial environment.

Necessity :

To overcome these problems, it is necessary to introduce a software-based monitoring and control system. Such a system will help in managing all activities from a single platform, making the process faster and more reliable. It will allow real-time monitoring of production, machines, and inventory, which will help the management take quick decisions.

Automation will reduce the need for manual work, which means fewer errors and less time spent on routine tasks. Employees will be able to focus more on important work instead of spending time on repetitive activities. The system will also improve communication between departments, as everyone will have access to the same updated information.

Another important reason for this system is better resource management. With proper data and analysis, the company can identify where resources are being wasted and take steps to improve efficiency. The

system can also help in predicting machine maintenance needs, which will reduce unexpected breakdowns.

Overall, the new system is necessary to make the industry more organized, efficient, and capable of handling future challenges.

Economic Value :

The proposed system will bring many economic benefits to the company. First, it will reduce the cost of manual labor by automating many tasks. At the same time, it will reduce errors, which often lead to financial losses.

With real-time monitoring, the company can quickly identify problems and solve them before they become serious. This will save both time and money. Better inventory management will also help in avoiding unnecessary expenses caused by overstocking or shortage of materials.

Energy efficiency is another major advantage. By tracking energy usage, the company can reduce waste and lower electricity costs. This is very important for an industry where energy consumption is high.

The system will also improve productivity by making operations smoother and faster. This means the company can produce more in less time, increasing its overall profit. Although the system will require some initial investment, the long-term benefits will be much greater.

Conclusion :

From our industrial visit, it is clear that Bangladesh Cable Shilpa Limited needs a more advanced and efficient system to manage its operations. The current manual and partially digital processes are not enough to meet modern industrial demands.

A software-based monitoring and control system will solve many existing problems by improving accuracy, saving time, and reducing costs. It will also help in better decision-making and resource management.

In conclusion, recognizing the need for such a system is an important step toward modernization. By implementing this system, the company can improve its performance and stay competitive in the long run.